

Task 2 – Structural Simulation (FEA)

Finite Element Analysis of a Simple Beam

1. Introduction

Finite Element Analysis (FEA) is a useful engineering method for studying how a component behaves when a load is applied. For this task, a simple beam was selected and analysed using FreeCAD. The purpose of the simulation was to study the stress distribution, deformation, and safety of the beam under the applied loading condition.

2. Objective

The objective was to perform a basic structural FEA on a simple beam. A mechanical load and boundary condition were applied to the model, after which the beam was meshed and solved using the CalculiX solver. The resulting stress and displacement were then examined and used to evaluate the structural response of the beam.

3. Model and Material

A simple rectangular beam was used as the component for the analysis. The beam was divided into small finite elements so that the solver could calculate its response at different locations. Steel St-E-255 was selected as the material for the beam. The material properties were assigned before the simulation was solved.

4. Boundary Conditions and Applied Load

One side of the beam was fixed to represent a securely supported end. This boundary condition prevents movement at the selected region. A force of 100 N was applied at the opposite end of the beam. These conditions were used to represent the loading case for the static structural analysis.

5. Meshing and Analysis Procedure

The beam was divided into a finite element mesh consisting of small triangular elements. After the mesh was created, the material, fixed support, and applied force were included in the analysis. The model was then solved using CalculiX. The solver completed the calculation successfully without an error, and the resulting stress and displacement fields were examined.

6. Stress Distribution

The von Mises stress result was used to evaluate the level of equivalent stress in the beam. The maximum von Mises stress obtained from the simulation was **1145.11 kPa**, which is approximately **1.145 MPa**. The highest stress was located close to the loaded region. This is reasonable because the applied force is transferred into the beam through this area.

7. Deformation

The displacement magnitude result shows how much the beam moves from its original position. The maximum displacement obtained was **102.66 nm**, which is approximately **0.00010266 mm**. The displacement is very small, indicating that the beam undergoes only a small amount of deformation under the 100 N load.

8. Factor of Safety Calculation

The factor of safety was estimated by comparing the material yield strength with the maximum von Mises stress. Using a yield strength of **255 MPa** for the selected St-E-255 material, the calculation is:

Factor of Safety = Yield Strength / Maximum von Mises Stress

FOS = 255 MPa / 1.14511 MPa

FOS ≈ 222.7

This gives a factor of safety of approximately **222.7**. This is a very high value, mainly because the applied load in this simulation produces a relatively low stress compared with the assumed yield strength.

9. Observations

The simulation shows that the beam remains stable under the selected loading condition. The highest stress is concentrated near the loaded region, while the maximum displacement occurs toward the loaded end. The calculated displacement is extremely small, which indicates that the beam is relatively stiff for the applied 100 N force. The maximum stress is also well below the assumed yield strength, giving a large calculated factor of safety.

10. Conclusion

The static structural FEA of the simple beam was successfully completed in FreeCAD using the FEM workbench and CalculiX solver. The beam was assigned Steel St-E-255 material, one side was fixed, and a 100 N force was applied to the opposite end. The maximum von Mises stress was 1145.11 kPa (1.145 MPa), while the maximum displacement was 102.66 nm. Using an assumed yield strength of 255 MPa, the calculated factor of safety was approximately 222.7. Overall, the results indicate low stress and very small deformation for the selected loading condition.

11. Final Remarks

This task provided practical experience with the main stages of a structural FEA: preparing the model, assigning material properties, applying boundary conditions and loads, generating a mesh, solving the model, and interpreting stress and deformation results. The simulation demonstrates how FEA can be used to evaluate the structural behaviour of a component before physical testing.